Coral Heat Stress Early Warning

Investigating Machine Learning Robustness Under Climate Change

Research Report

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Field: Artificial Intelligence and Marine Conservation

Overview

Climate change creates a major challenge for environmental prediction systems: future conditions may differ significantly from historical observations. This project investigates whether satellite-derived sea surface temperature (SST) data can provide early warning signals of coral heat stress and examines how predictive models behave under climate-induced distribution shift.

Research Questions

- Can short-term coral thermal stress risk be detected using SST observations?
- How reliable are predictions when environmental conditions change?
- Do models learn meaningful stress dynamics or rely on historical temperature thresholds?

Data and Methodology

The study uses NOAA Coral Reef Watch SST data. The dataset was expanded from approximately one year to six years of observations to improve seasonal coverage and capture more thermal stress events.

The research pipeline included:

- SST processing and anomaly calculation
- Temporal feature engineering
- Future anomaly prediction
- Time-based validation without random shuffling
- XGBoost classification modelling

Methodological Discovery

Early experiments produced ROC-AUC values around 0.65. Further analysis revealed issues including inconsistent pipelines and possible target leakage. After restructuring, the baseline performance decreased, showing that reliable evaluation is more important than optimistic metrics.

The prediction task was redesigned to avoid using target information inside the input features. This created a more realistic early-warning problem.

Final Results

After improving the dataset and methodology:

Current conditions: ROC-AUC ≈ 0.91

+0.5°C warming: ROC-AUC ≈ 0.89

+1°C warming: ROC-AUC ≈ 0.84

+2°C warming: ROC-AUC ≈ 0.60

Distribution Shift Finding

The model remains robust under moderate warming but experiences significant degradation under extreme temperature shifts.This indicates that the model captures meaningful patterns such as seasonal cycles, thermal accumulation, and short-term temperature dynamics. However, it also relies partly on absolute SST thresholds, which may become unreliable as ocean temperatures increase.

Scientific Significance

The key finding is that environmental AI systems must be evaluated not only for accuracy under current conditions, but also for robustness under future climate scenarios. A model that performs well today may fail when the environment changes.

Future Research

Future work includes:

- Testing across multiple reef locations
- Evaluating future-year generalization
- Exploring climate-invariant features
- Improving uncertainty estimation

Conclusion

This project evolved from a prediction task into an investigation of how machine learning systems behave under climate change. The research highlights